

Qatar-1b

R-2782

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130812 · created 2026-09-04T00:04:41+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456516.688114 +/- 9.8e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.136 +/- 0.016
Transit depth (Rp/Rs)^2	1.86 +/- 0.44 [%]
Orbital Inclination (inc)	89.03 +/- 1.2
Airmass coefficient 1 (a1)	1.132 +/- 0.022
Airmass coefficient 2 (a2)	-0.1 +/- 0.044
Scatter in the residuals of the lightcurve fit is	1.91 %
Best Comparison Star	#2 - [557, 191]
Optimal Aperture	2.65
Optimal Annulus	13.18
Transit Duration (day)	0.0818 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.136 $\pm$ 0.016 vs 0.1463 (deviation 0.41 σ)
Duration fit vs published	0.0818 d vs 0.0692 d (deviation 4.77 σ)
Mid-time O-C	0.2 min
Depth SNR	5.5
Residual scatter	1.91 %

Light curve

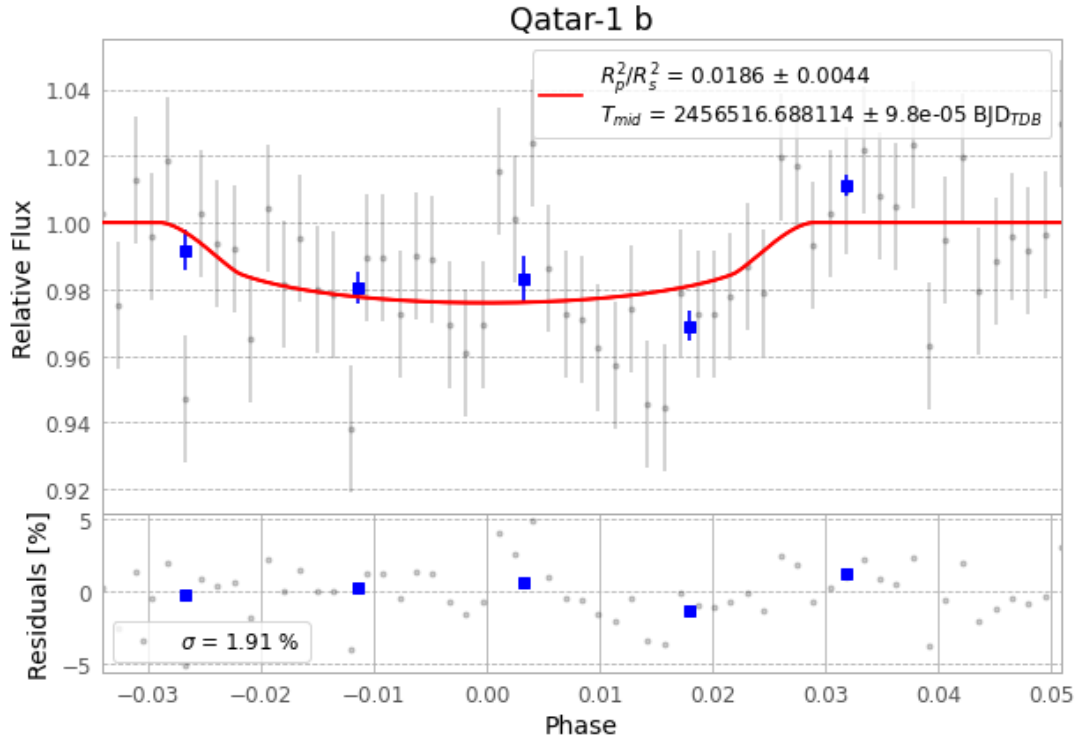


Figure 1 — FinalLightCurve_Qatar-1 b_2013-08-11.png (SHA-256 d0cbd3fe43165e0e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [509, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c7c846dcffd72f7a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), Qatar-1130812031813.FITS → Qatar-1130812061813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130812.log (e8898417762d...), FinalLightCurve_Qatar-1 b_2013-08-11.png (d0cbd3fe4316...), FinalLightCurve_Qatar-1 b_2013-08-11.csv (42d90db499dc...)

Compute resources

Wall time 126.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:04Z.